VOLT CLOUD

Modular Audio Environment

User Manual · Version 3.0

A browser-based modular synthesizer and generative audio environment built on the Web Audio API. Connect modules with virtual cables, sequence, process, and generate sound — no plugins required.

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1. GETTING STARTED

What is Volt Cloud?

Volt Cloud is a modular audio environment for Windows (and browser). Build signal chains — called patches — by connecting modules with virtual cables, just like a hardware Eurorack modular synthesizer. Audio processing runs entirely via the Web Audio API. No plugins, no internet connection required after installation.

Activation (first launch)

On first launch, Volt Cloud displays an activation window. Enter your license key and click ACTIVATE. Keys are in the format `VC-XXXX-XXXX-XXXX`. Once activated, the key is saved permanently — you will not be asked again on subsequent launches.

To purchase a license: voltcloud.gumroad.com

To reset activation (e.g. moving to a new PC): delete `%APPDATA%\voltcloud\license.json` *and relaunch.*

System Requirements

OS	Windows 10 / 11 (64-bit)
Audio	Any audio output device; ASIO optional
MIDI	Any class-compliant USB MIDI controller
RAM	4 GB minimum, 8 GB recommended
CPU	Modern multi-core processor (Intel i5 / Ryzen 5 or better)

2. INTERFACE OVERVIEW

The Canvas

The main area where you place and connect modules. Drag on empty space to move modules. The canvas expands infinitely — modules placed off-screen remain accessible by scrolling.

The Palette (Left Sidebar)

Lists all available modules organized by category. Click any name to add it to the canvas. Categories:

Synthesis	Analog-X, West Coast Synth, Oscillator, VCO 2600
Generative Synthesis	Mist, Spectrum, GhostGrid — self-generating sound engines
Sampling	Granular, Swarm, File Loader, Drift, PD Patch
CV Sequencers	Gen Sequencer, Turing Machine, CV Sequencer
CV Utilities	LFO, ADSR, WASP VCF, CV Mixer, Crossfader, 5-Band EQ, Gain, Compressor, Limiter
Effects	Reverb, Haze, Delay, Chorus, Flanger, Phaser, Distortion, Bitcrusher, Tremolo, Ring Mod, Panner
Output	Master Out

Cables

Click an output port, then click an input port to connect. Right-click a cable or click it and press Delete to remove. Audio cables are color-coded; CV (modulation) cables are amber.

Themes

Three themes available via the BRIGHT / MEDIUM / DARK buttons in the sidebar: Bright (parchment/beige), Medium (warm grey), Dark (near-black). Theme preference is not saved between sessions.

Save / Load Patches

▼ **SAVE PATCH** — opens a name dialog, then downloads a `.voltcloud` file. You can choose to embed audio files (Granular samples, captured audio) and PD Patch files. Embedded patches are larger but fully portable — share the single file with anyone.

▲ **LOAD PATCH** — opens a file picker. Restores all modules, cables, and parameters exactly.

⬧ VIEW AS PD

Shows a real-time Pure Data patch diagram of your current Volt Cloud patch — rendered in authentic Pd style (white canvas, monospace fonts, Windows-grey title bar). Each module maps to its closest Pd equivalent objects with patch cords drawn between them.

3. MIDI SYSTEM

Volt Cloud uses the Web MIDI API, which works in Electron (the desktop app) and Chrome/Edge browsers. No additional drivers are needed for most devices. Korg controllers: uninstall the Korg USB-MIDI driver and use the generic Windows class-compliant driver.

Connecting Devices

Click ■ **CONNECT MIDI** in the palette sidebar. The browser requests MIDI permission — click Allow. All connected devices appear as checkboxes. **All checked = all devices active.** Uncheck any device to mute it from the global MIDI dispatch.

Per-Module Device Routing

Every MIDI-enabled synth module has a **device dropdown** below its ■ MIDI button. Set it to "Any device" (default) or pick a specific controller. This lets you route different controllers to different modules simultaneously — e.g. NanoKey2 → Analog-X, Launchpad → GhostGrid.

■ MIDI ON / OFF Toggle

Every MIDI-receiving module has a MIDI toggle button. Green = MIDI enabled (responds to controller). Red/orange = MIDI disabled (module runs as standalone generator/effect, ignores keyboard input). Modules with MIDI toggle: Analog-X, West Coast Synth, Mist, Spectrum, GhostGrid, ADSR, Granular.

Computer Keyboard

A–L = white keys (C to E). W/E/T/Y/U/O/P = black keys. Z = octave down, X = octave up. Works for all MIDI-enabled synth modules simultaneously.

4. SYNTHESIS MODULES

ANALOG-X

Roland Juno-X inspired 6-voice polyphonic synthesizer

DCO Shapes	SAW, PULSE (variable width), SUB (oct below), SSAW (7-voice supersaw)
VCF	24dB resonant lowpass + passive highpass. Resonance self-oscillates at high values
Envelopes	Three independent: Filter ADSR, Amp ADSR, Pitch ADSR
LFO	Sine/Tri/Saw/Square/S+H. Routes to PITCH, FILT, AMP, PW simultaneously
Chorus	OFF / I (slow) / II (classic Juno) / III (fast lush)
Octave	-4 to +4 octave transpose via - / + buttons
MIDI toggle	Green = responds to controller. Orange = standalone. Device dropdown available
Presets	20 factory presets including JP Strings, Dream Pad, Hoovery, Sub Reso Bass

Connect a MIDI keyboard or use computer keyboard (A-L = white keys). Polyphonic — 6 simultaneous voices.

WEST COAST SYNTH

Buchla-inspired synthesis with wavefolder and Lowpass Gate

Primary Osc	Frequency, waveform selection, wavefold amount
Modulator	FM modulator oscillator for complex timbres
LPG	Lowpass Gate — TRIG (full velocity) / SOFT (gentle) buttons for envelope
Reverb	Built-in spring reverb with mix control
MIDI toggle	Green = responds to controller. Device dropdown available
RND	Randomizes all parameters for instant new timbre

TRIG button fires a full-velocity gate. SOFT button fires a gentle gate. Connect CV to trig-cv for sequenced triggering.

VCO 2600

ARP 2600-inspired voltage-controlled oscillator

Coarse	Frequency 10Hz-10kHz
Fine Tune	+/-100 cents
LF MODE	Switch to 0.03-30Hz for LFO use
Pulse Width	10%-90% (Pulse output only)
Outputs	Sine, Triangle, Sawtooth, Pulse — all four simultaneous and independent
Inputs	freq-cv (1V/oct), fm1-cv (exponential FM), fm2-cv (linear FM), pwm-cv

All four outputs are active simultaneously. Connect each to a different destination for layered timbres.

OSCILLATOR

Simple single-waveform oscillator

Waveform	Sine, Sawtooth, Square, Triangle
Frequency	Continuous 20Hz-20kHz

Useful as a simple tone generator, test oscillator, or LFO source. Pair with LFO for vibrato.

5. GENERATIVE SYNTHESIS

These modules generate music continuously without external input. Patch them directly to Master Out and press START or EVOLVE.

MIST

Brian Eno Bloom-inspired generative ambient synthesizer

Tone	Sine, FM Bell, Triangle, Sawtooth, Square, Organ, Pluck, Glass — each distinctly different
Scale / Root	Musical scale and root note for all generated tones
Delay	Time between automatic blooms (lower = busier)
Decay	How quickly each tone fades
Ring	Sustain character — how long tones ring
Pan Spread	Stereo width of random panning
■ EVOLVE	Starts continuous self-generation. Tones fire automatically and overlap organically
■ RND	Randomizes all parameters for a new character
MIDI toggle	ON = responds to keyboard and plays notes. OFF = fully autonomous generator

Press EVOLVE and patch to Master Out for instant ambient music. Delay controls the density of generation.

SPECTRUM

Spectral arpeggiator with 4 independently timed sectors

■ START	Begins continuous tone generation across all active sectors
Sectors 1-4	Each sector fires notes in its frequency range independently. Toggle each on/off
Attack / Decay	Envelope shape per sector
Ring	Sustain amount (capped at 0.82 to prevent feedback)
Scale / Root	Musical scale for all generated notes
Timescale	Overall speed multiplier
■ RND	Randomizes all sector parameters
Tone	Pure sine voice — no resonant filters to prevent feedback
Note range	Capped at 2 octaves above root, maximum MIDI 72 (C5) to prevent harsh high notes

Spectrum produces layered evolving tones. If you hear feedback, lower the Ring values on each sector.

GHOST GRID

Generative Launchpad synthesizer — 8x8 grid mapped to musical scale

Grid	8x8 canvas. Rows = octaves, columns = scale degrees. Every pad is always a correct note
Pad tap	Plants a seed: plays a note + spawns a 3-7 note generative sequence that fades over time
Bottom row	Drone layer — tap any bottom pad to toggle root+fifth sustained drone
Drone Vol/Oct	Controls drone volume and octave offset (-4 to +4)

Drone Interval	Root+5th, Root+4th, Root+Oct, Root+Maj3rd, Root+Min3rd, Root only
Decay / Drift	How quickly sequences fade; probability of note mutation each loop
Interact	Neighboring sequences nudge each other's timing (emergent polyrhythm)
Scale	16 scales: Pentatonic, Major, Minor, Dorian, Whole Tone, Chromatic, Hirajoshi, Phrygian, Blues, Mixolydian, Locriar
Octave	Drone octave: -4 to +4
⊕ LAUNCHPAD	Connects directly to Launchpad MK2 via Web MIDI for hardware control and LED feedback
LED feedback	Pads glow by column color while active, fade as sequences decay. Bottom row glows amber for drone
MIDI toggle	ON = also responds to MIDI note input. Device dropdown selects which controller
■ RND	Plants 4-8 random seeds to get things going

The Launchpad bottom row toggles the drone from hardware. Connect GhostGrid output to Master Out then press START.

6. SAMPLING MODULES

GRANULAR

Granular synthesis engine — slice and scatter any audio

LOAD SAMPLE	Load any audio file (MP3, WAV, OGG, FLAC). File embeds in saved patches
■ GRAB	Capture audio from the input port into the grain buffer (4 second window)
[REC] LIVE	Continuously captures from input port every 400ms for real-time granulation
■ START	Begins grain playback
Position	Playhead position in the buffer (0=start, 1=end)
Size	Grain duration 5ms-800ms
Density	Grains per second (hard-capped at 15/s to prevent overload)
Scatter	Random position offset per grain
Pitch	Pitch multiplier (0.25-4x)
Feedback	Feed output back into grain buffer for cascading effects
MIDI toggle	ON = MIDI pitch control. OFF = standalone texture generator
Device select	Choose which MIDI controller drives pitch

Connect Mist or any synth to the audio input port, then press GRAB to capture it as a grain buffer. Granular crashes can occur if connected to Master Out with no audio — always load a sample or grab audio first.

SWARM

Multi-voice drone synthesizer — up to 28 simultaneous grains

Voices	Number of simultaneous oscillators
Spread	Frequency spread across voices
Detune	Random pitch offset per voice

DRIFT

Micro-detuned oscillator cluster for warm pad textures

Voices	2-8 oscillators
Rate	Drift/wander speed
Depth	Maximum detune amount

7. CV UTILITIES & MIXING

CROSSFADER

DJ-style 2-channel audio crossfader

CH A / CH B	Two independent audio inputs
Fader	Slide left = 100% A, slide right = 100% B, center = both at 70% (equal-power curve)
Level meters	Show current gain for each channel in real time
Output	Single mixed audio output with built-in limiter to prevent clipping

Uses equal-power (cosine) crossfade — no volume dip in the center position, same as professional DJ mixers.

5-BAND EQ

Parametric equalizer — 5 bands from sub to air

SUB	Low shelf at 80Hz. Controls sub-bass weight
LOW	Peaking band at 300Hz. Controls low-mid muddiness or warmth
MID	Peaking band at 1kHz. Controls presence and definition
HIGH	Peaking band at 4kHz. Controls clarity and edge
AIR	High shelf at 10kHz. Controls shimmer and brightness
Gain range	+/-15 dB per band. Vertical fader — center = 0dB (flat)
ACTIVE/BYPASS	Bypasses all bands (sets to 0dB) without losing settings
FLAT	Resets all bands to 0dB immediately
Output	Master output level 0-2x

Chain after any synth or effect module for tonal shaping. SUB and AIR are shelving filters; LOW, MID, HIGH are peaking (bell) filters.

ADSR ENVELOPE

Triggered amplitude envelope generator

Attack	1ms-10s rise time
Decay	1ms-10s fall to sustain level
Sustain	0-1 level held while gate is open
Release	1ms-20s fall after gate closes
MIDI toggle	ON = triggered by incoming MIDI notes. Device dropdown available

LFO

Low-frequency oscillator for modulation

Rate	0.01-20Hz (or lower in LF mode)
Depth	Output amplitude

Shape	Sine, Triangle, Sawtooth, Square, S+H (Sample & Hold)

Connect LFO output to filter cutoff CV, pitch CV, or any other CV input for modulation.

LINE IN

Live audio input from soundcard or audio interface

Device	Select from all available audio inputs on your system
Input Gain	0–2× amplification of incoming signal
VU Meter	Real-time level display with clip indicator (lights red on overload)
START / STOP	Request audio permission and open the input stream
MUTE	Cut output without stopping the stream — zero-latency silencing
M (Mono)	Force mono sum from stereo input

Disable echo cancellation, noise suppression, and auto-gain in your OS audio settings for the cleanest signal. Connect LINE IN output to Granular for live texture capture, or to any effect chain.

WASM PLUGIN HOST

Load WAM2 and Faust WASM plugins as Volt Cloud modules

▲ FOLDER	Load a WAM2 plugin from a local folder — any webaudiomodules.com plugin
▲ ZIP	Load a WAM2 or Faust IDE export from a .zip file
▲ FAUST	Load a raw Faust .wasm + .json file pair directly
Plugin GUI	Renders the plugin native GUI if available; auto-generates parameter sliders as fallback
In Gain / Out Gain	Pre-plugin and post-plugin level control
BYPASS	Hard bypass — routes audio around the plugin with zero latency
ACTIVE / BYPASS toggle	Switch plugin in and out of signal chain
MIDI Routing	Route global MIDI events to the plugin with optional per-channel filtering
PARAMS toggle	Switch between native GUI and auto-generated parameter slider view
SAVE / LOAD preset	Snapshot and restore all parameter values
✕ Unload	Remove current plugin and free all memory

Getting WASM Plugins

WAM2 plugins: visit webaudiomodules.com and download any plugin folder or .zip. Load via ▲ FOLDER or ▲ ZIP. Hundreds of free effects, synths, and processors available. Works completely offline after download.

wam-community (recommended starting point): clone or download github.com/boourns/wam-community — a curated collection of over 40 prebuilt WAM2 plugins ready to use with no build step required. Each plugin is a folder containing an index.js and supporting files. Open the folder for the plugin you want and load it via ▲ FOLDER. Includes effects (reverb, delay, chorus, distortion, EQ, filter, compressor, bitcrusher), synthesizers (FM synth, organ, sampler, wavetable), MIDI processors, and sequencers. All work offline once downloaded.

Faust IDE export: go to faustide.grame.fr in any browser. Load any of the hundreds of built-in DSP examples (reverbs, filters, pitch shifters, physical models, granular effects, vocoders) or write your own Faust DSP code. Click Export → Platform: Web → Architecture: FAUSTwasm. Download the .zip. Load via ▲ ZIP. Works offline from that moment on.

Raw Faust WASM: if you compile Faust yourself using the command-line tools or faustwasm npm package, select both the .wasm file and the accompanying .json metadata file using the ▲ FAUST button.

8. EFFECTS

REVERB / ROOM VERB / CLOUDS REVERB

Convolution reverb with multiple room sizes and character. Clouds Reverb adds granular shimmer.

HAZE

Tape-style delay with flutter, saturation, and frequency drift. Creates vintage lo-fi echo. Uses theme colors.

DELAY

Clean digital delay with feedback and mix control.

CHORUS

BBD-style chorus with 3 modulated delay voices for width and warmth.

FLANGER

Short delay with feedback swept by an LFO for the classic jet-plane sweep.

PHASER

All-pass filter chain swept by LFO for swooping phase effects.

DISTORTION

Three modes: Soft Clip (warm overdrive), Hard Clip (aggressive), Wave Fold (complex harmonics). Pre-gain and post-gain control.

BITCRUSHER

Reduces bit depth (1-16 bit) and sample rate for lo-fi digital crunch.

RING MOD

Ring modulator — multiplies input by a carrier oscillator for metallic/robotic textures.

TREMOLO

Amplitude modulation — volume fluctuation at LFO rate. Sine, Square, Saw, Triangle shapes.

PANNER

Auto-panner with LFO-controlled stereo position.

9. GHOST GRID + LAUNCHPAD SETUP

GhostGrid has dedicated Launchpad MK2 integration — the hardware pads control the module and LEDs mirror the active sequences in real time.

1. Add GhostGrid module from the Generative Synthesis category
2. Connect its AUDIO output to Master Out
3. Connect MIDI — click ■ CONNECT MIDI in the sidebar, allow browser permission
4. In GhostGrid, press ⊕ LAUNCHPAD — status bar shows "✓ Launchpad MK2"
5. The Launchpad flashes purple to confirm connection
6. Press any pad on the Launchpad — a seed sequence begins and the pad lights up
7. Bottom row (row 8) toggles the drone on/off from hardware
8. Select scale and root from the dropdowns. All pads play in-scale notes
9. Press ■ RND to auto-seed 4-8 random pads and get things going immediately
10. Adjust Decay, Drift, and Interact to shape how sequences evolve over time

LED Colors on Launchpad:

White flash	Seed event (pad was just triggered)
Green (bright)	Active sequence, young / full volume
Column color	Each column has its own hue — sequences keep their column color as they age
Indigo (dim)	Sequence fading out near end of life
Amber pulse	Drone row — slow breathing pulse when drone is on
Dim white dots	Root notes shown faintly on dormant pads so you can see the scale layout

10. TIPS & TECHNIQUES

Instant ambient patch

Add Mist → Master Out. Press EVOLVE. Done. Adjust Delay for density, Ring for sustain, Scale/Root for tonality. Add Room Verb between Mist and Master Out for depth.

Granular texture from synth

Add Mist → Granular (audio input) → Master Out. Press EVOLVE on Mist to start it. In Granular, press GRAB to capture 4 seconds of Mist, then press START. The granular engine will scatter Mist fragments across time.

DJ crossfade between two synths

Add Analog-X → Crossfader CH A. Add Spectrum (running) → Crossfader CH B. Connect Crossfader output → Master Out. Slide the fader to blend between them.

EQ placement

Place 5-Band EQ after any synth and before reverb. Cut SUB on Mist/Spectrum to remove rumble. Boost AIR slightly on Analog-X pads for presence.

GhostGrid generative performance

Run GhostGrid with Launchpad. Set scale to Hirajoshi or Pentatonic. Tap a few pads in the upper rows for melody, then hold a bottom pad for drone. Let sequences evolve, occasionally seeding new pads to refresh the texture.

Saving embedded patches

When saving, choose to embed both audio and PD files. The resulting .voltcloud file contains everything — send it to collaborators and they get the full patch including any recorded audio and PD patches.

Prevent Granular overload

Keep Density below 12/s. Do not connect Granular to Master Out before loading a sample or grabbing audio. The module has a built-in grain limit of 12 simultaneous voices.

Multiple MIDI controllers

Connect MIDI. Each synth module has a device dropdown — assign NanoKey2 to Analog-X and Launchpad to GhostGrid. Both controllers work simultaneously and independently.

11. TROUBLESHOOTING

No sound at all

Check that Master Out is connected and volume is up. Click anywhere on the canvas to unlock the AudioContext (browser security requirement). Verify your system audio output device is active.

MIDI devices not appearing

Click ■ CONNECT MIDI and allow the browser MIDI permission popup. For Korg devices: uninstall the Korg USB-MIDI driver, use Windows generic driver instead.

MIDI dropdown is empty after connecting

The dropdown populates automatically when MIDI connects. If empty, click CONNECT MIDI again while the device is plugged in.

App crashes (0xC0000005)

This is a Windows audio driver memory error. Check the crash log at %APPDATA%\voltcloud\voltcloud-crash.log. Most common cause: connecting Granular to Master Out before loading a sample.

Spectrum producing high-pitched ringing

Lower the Ring values on each sector (Spectrum UI). Ring above 0.82 causes notes to sustain indefinitely and stack. Use the ■ RND button which caps ring at 0.4 automatically.

Mist not generating tones with EVOLVE

Press EVOLVE — the button turns green and shows ■ EVOLVE. If no sound, click the canvas first to unlock the AudioContext, then press EVOLVE again.

Save dialog shows an error

The save dialog requires typing a patch name. If you see a prompt() error, update to the latest version which uses a custom dialog.

License activation fails

Keys are case-insensitive and dashes are optional. Try typing the key without dashes. To reset: delete %APPDATA%\voltcloud\license.json and relaunch.

GhostGrid Launchpad not connecting

Press ⊕ LAUNCHPAD in the GhostGrid module specifically — this requests its own MIDI access with SysEx permission. The status bar will show the device name when connected. A purple sweep flash on the Launchpad confirms the connection.

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Support: voltcloud.gumroad.com
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